

Question number	Answer	Mark
4(a)	An answer that includes the following points: • changes to {weather patterns / global temperatures} over a long period of time (1) • due to human activity (1)	(2)

Question number	Answer	Additional guidance	Mark
4(b)(i)	Any two from: • carbon dioxide • methane • water vapour • nitrogen oxides • CFCs	Accept correct chemical formulae	(1)

Question number	Answer	Mark
4(b)(ii)	An explanation that includes the following points: • accumulated gases in the atmosphere trap infrared radiation (1) • resulting in an increase in the temperature of the Earth's {atmosphere / surface} (1)	(2)

Question number	Answer	Mark
4(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • carbon dioxide is released into atmosphere • because agriculture requires farm machinery powered by burning fossil fuels • because industry burns fossil fuels for energy • because homes and business buildings use fossil fuels for heating • transport uses fossil fuels • fossil fuels are burnt to produce electricity • forestry may decrease the number of trees • so less carbon dioxide removed from air by photosynthesis	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific